Quadratic Equation

$\Rightarrow ax^2 + bx + c \Rightarrow 0 \; ; \; a \neq 0$

Note: Quadratic Eqⁿ has 2 roots always.

Eg $x^2 + 5x - 6 \Rightarrow 0$

$x^2 + 6x - x - 6 \Rightarrow 0$

$x(x+6) - (x+6) \Rightarrow 0$

$(x+6)(x-1) \Rightarrow 0$

$x \Rightarrow -6$ and 1

∴ The method that We Used here in factorization ⇒ Cause if $a \Rightarrow 0$ {becomes} then it will become an Linear Eqⁿ, but to make it Quadratic it has to satisfy the Condition $a \neq 0$.

Roots of Quadratic Equation

The roots of the quadratic Eqⁿ $\Rightarrow ax^2 + bx + c \doteq 0$ is given

$$x \Rightarrow \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

∴ The Expression $D \Rightarrow b^2 - 4ac$ is Called the discriminant of the Quadratic Eqⁿ.

Equation Convertible to Quadratic form

Ex) $x^4 - 5x^2 + 6 \Rightarrow 0$

→ This is not an Quadratic Equation;
but to make it Quadratic Let's Assume

$\therefore x^2 \Rightarrow t$

$t^2 - 5t + 6 \Rightarrow 0$ { Quadratic Eq }

$t^2 - 3t - 2t + 6 \Rightarrow 0$

$t(t-3) - 2(t-3) \Rightarrow 0$

$(t-3)(t-2) \Rightarrow 0$

$t \Rightarrow 3$ and 2

$x^2 \Rightarrow 3$ and $x^2 \Rightarrow 2$

$x \Rightarrow \pm\sqrt{3}$ and $x \Rightarrow \pm\sqrt{2}$

Ex) The roots of the Eq $4^x - 3.2^{x+3} + 128 \Rightarrow 0$

$\Rightarrow 4^x - 3.2^{x+3} + 128 \Rightarrow 0$

Assume,

$(2^2)^x - 3.2^x.2^3 + 128 \Rightarrow 0$

$(2^x)^2 - 3.2^x.2^3 + 128 \Rightarrow 0$

$2^x \Rightarrow t$

$(t)^2 - 3t2^3 + 128 \Rightarrow 0$

$t^2 - 24t + 128 \Rightarrow 0$ { $t^2 - 16t - 8t + 128 \Rightarrow 0$ }

$t(t-16) - 8(t-16) \Rightarrow 0$

$(t-16)(t-8) \Rightarrow 0$

$\therefore t \Rightarrow \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$D \Rightarrow (24)^2 - 4(1)(128)$

$\Rightarrow 576 - 512$

$\Rightarrow 64$

$$t \Rightarrow \frac{-b \pm \sqrt{64}}{2a}$$

$$t \Rightarrow \frac{-b + \sqrt{64}}{2a} \text{ and } \frac{-b - \sqrt{64}}{2a}$$

$b \Rightarrow -24$, $a \Rightarrow 1$ an

$$t \Rightarrow \frac{-(-24) + 8}{2} \text{ and } \frac{-(-24) - 8}{2}$$

$$t \Rightarrow \frac{+16}{2} \Rightarrow +8 \text{ and } \frac{+32}{2} \Rightarrow +16$$

$\Rightarrow t \Rightarrow +8$ and $+16$

$\therefore t \Rightarrow 2^x$

$2^x \Rightarrow +8$ and $+16$

$2^x \Rightarrow 2^3$ and 2^4.

$\Rightarrow x \Rightarrow 3$ and 4

Ex) find the Real Roots of the Eq^n

$(x-1)(x-2)(x-3)(x-4) \Rightarrow 3$.

$\Rightarrow$ Note:- It is simply Sol^n to an Eq^n that is a real number (not an imaginary/complex number).

$x \Rightarrow \pm 3$ (real number)

$x \Rightarrow \pm i$ (imaginary)

$\rightarrow (x-1)(x-2)(x-3)(x-4) \Rightarrow 3$

$x - 1 \Rightarrow 3$, $x \Rightarrow 4$ $\{(4-1)(4-2)(4-3)(4-4) \Rightarrow 3\}$

$x - 2 \Rightarrow 3$, $x \Rightarrow 5$

$x - 3 \Rightarrow 3$, $x \Rightarrow 6$

$x - 4 \Rightarrow 3$, $x \Rightarrow 7$

{ Wrong method }

In this type of Question Try to always Look for Substitution.

→ $(x-1)(x-2)(x-3)(x-4) \Rightarrow 3$

$(x^2-2x-x+2)(x^2-4x-3x+12) \Rightarrow 3$

$(x^2-3x+2)(x^2-7x+12) \Rightarrow 3$

$x^2-3x+2 \Rightarrow 3$ | $x^2-7x+12 \Rightarrow 3$ {Wrong}

$x^2-3x-1 \Rightarrow 0$ | $x^2-7x+9 \Rightarrow 0$

→ Long approach

$\Rightarrow (x-1)(x-2)(x-3)(x-4) \Rightarrow 3$

→ $(x-1)(x-4)(x-2)(x-3) \Rightarrow 3$

→ $(x^2-5x+4)(x^2-5x+6) \Rightarrow 3$

→ $(x^2-5x+4)((x^2-5x+4)+2) \Rightarrow 3$

$\Rightarrow \therefore x^2-5x+4 \Rightarrow t \quad -(1)$

→ $t(t+2) \Rightarrow 3$

$t^2+2t-3 \Rightarrow 0$

$t^2+3t-t-3 \Rightarrow 0$

$t(t+3)-1(t+3) \Rightarrow 0$

$(t+3)(t-1) \Rightarrow 0$

$\Rightarrow t \Rightarrow -3$ and 1

→ Substitute this value in (1)

$x^2-5x+4 \Rightarrow -3$ and $x^2-5x+4 \Rightarrow 1$

$x^2-5x+7 \Rightarrow 0$ | $x^2-5x+3 \Rightarrow 0$

$x \Rightarrow \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ | $x \Rightarrow \frac{-b+\sqrt{b^2-4ac}}{2a}$

$\Rightarrow \frac{-(-5) \pm \sqrt{(-5)^2+4(1)(7)}}{2(1)}$ | $x \Rightarrow \frac{-(-5) \pm \sqrt{(-5)^2}}{2(1)}$

$\Rightarrow$ not Real Roots | $x \Rightarrow \frac{5+\sqrt{13}}{2}$

Methot ⇒ 2

We Can Say

$(x-1)(x-2) \Rightarrow t$

then automatically,

$(x-3)(x-4) \Rightarrow \frac{3}{t}$

$\left\{ \begin{array}{l} A \Rightarrow t, \; B \Rightarrow \frac{3}{t} \\ A \cdot B \Rightarrow 3 \end{array} \right\}$

∴ But this will be more Complicated.

Ex) Let $\frac{-\pi}{6} < \theta < \frac{-\pi}{12}$, Suppose α_1 and β_1 are the roots of the eq^n $x^2 - 2x\sec\theta + 1 \Rightarrow 0$ and α_2 and β_2 are the roots of the eq^n $x^2 + 2x\tan\theta - 1 \Rightarrow 0$. if $\alpha_2 > \beta_2$ and $\alpha_1 > \beta_1$ then $\alpha_1 + \beta_2$ equals.

$\Rightarrow x^2 - 2x\sec\theta + 1 \Rightarrow 0$ and $x^2 + 2x\tan\theta - 1 \Rightarrow 0$

$\rightarrow x \Rightarrow \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x \Rightarrow \frac{2\sec\theta \pm \sqrt{4\sec^2\theta - 4}}{2}$

$x \Rightarrow \sec\theta \pm \sqrt{\sec^2\theta - 1}$

$x \Rightarrow \sec\theta \pm \sqrt{(\sec\theta)^2 - (1)^2}$

$x \Rightarrow \sec\theta \pm (\sec\theta - 1)$

$x \Rightarrow \sec\theta \pm (\sec\theta - 1)$

$x \Rightarrow \sec\theta \pm |\tan\theta|$

$x \Rightarrow \sec\theta + \tan\theta$ and $\sec\theta - \tan\theta$

$x \Rightarrow \frac{-2x\tan\theta \pm \sqrt{4x^2\tan^2\theta + 4}}{2(1)}$

$x \Rightarrow \frac{-2(x\tan\theta) \pm 2\sqrt{x^2\tan^2\theta + 1}}{2}$

$x \Rightarrow \frac{2(-x\tan\theta \pm \sqrt{x^2\tan^2\theta + 1})}{2}$

$x \Rightarrow -x\tan\theta \pm \sqrt{x^2\tan^2\theta + 1}$

$x = -x\tan\theta \pm |\sec\theta|$

$x \Rightarrow -\tan\theta + \sec\theta$ and $-\tan\theta - \sec\theta$

Make sure about the Quadrat. You have to think about if you substitute the value by an assumption which one will be greater and which won't

$\beta_1 \Rightarrow \sec\theta + \tan\theta$ | $\alpha \Rightarrow \sec\theta - \tan\theta + \sec\theta$ (+) [By seeing the domain]

$\alpha_1 \Rightarrow \sec\theta - \tan\theta$ | $\beta \Rightarrow -\tan\theta - \sec\theta$

$t \Rightarrow -2, 3$

$-(-2) + 3 \Rightarrow 2+3 \Rightarrow 5$

$-(-2) - 3 \Rightarrow 2-3 \Rightarrow -1$

We know;

$\rightarrow \alpha_1 > \beta_1$ and $\alpha_2 > \beta_2$

$\Rightarrow \alpha_1 + \beta_2$

$\Rightarrow \sec\theta + \tan\theta - \tan\theta - \sec\theta \Rightarrow 0$

$\Rightarrow \alpha_1 + \beta_2 \Rightarrow 0$.

$\alpha_1 \Rightarrow \sec\theta - \tan\theta$ | $\alpha_2 \Rightarrow -\tan\theta + \sec\theta$

$\beta_1 \Rightarrow \sec\theta + \tan\theta$ | $\beta_2 \Rightarrow -\tan\theta - \sec\theta$

$\Rightarrow \alpha_1 + \beta_2$

$\rightarrow \sec\theta - \tan\theta - \tan\theta - \sec\theta \Rightarrow 0$

$\Rightarrow \alpha_1 + \beta_2 \Rightarrow 0$

$\rightarrow \sec\theta - \tan\theta - \tan\theta - \sec\theta \Rightarrow -2\tan\theta$

Equation v/s Identity

→ $ax^2 + bx + c \Rightarrow 0$

If it has three Distinct Roots :- α, β and γ

∴→ roots means if you substitute them in the eqⁿ they will come out to be zero.

→ $a\alpha^2 + b\alpha + c \Rightarrow 0$ — ①
→ $a\beta^2 + b\beta + c \Rightarrow 0$ — ②
→ $a\gamma^2 + b\gamma + c \Rightarrow 0$ — ③

⇒ ① − ②
⇒ $a\alpha^2 + b\alpha + c - a\beta^2 - b\beta + c \Rightarrow 0$
$a\alpha^2 - a\beta^2 + b\alpha - b\beta \Rightarrow 0$
$a(\alpha^2 - \beta^2) + b(\alpha - \beta) \Rightarrow 0$
$a(\alpha - \beta)(\alpha + \beta) + b(\alpha - \beta) \Rightarrow 0$
$(\alpha - \beta)\,(a(\alpha + \beta) + b) \Rightarrow 0$

∴ for this value to become zero we have to think that what should we assume that this Eqⁿ becomes zero. We know $(\alpha - \beta)$ can never be zero cause the both are distinct values. so, We don't have any othe choice but to go and assume that the $(a(\alpha + \beta) + b) \Rightarrow 0$

So,

$a(\alpha + \beta) + b \Rightarrow 0$ — ④

(2) - (3)

→ $a\beta^2 + b\beta + c - ar^2 - br - c = 0$

→ $a\beta^2 - ar^2 + b\beta - br = 0$

→ $a(\beta^2 - r^2) + b(\beta - r) = 0$

→ $\beta - r\,(a(\beta + r) + b) = 0$

↳ Distinct

$a(\beta + r) + b = 0$ ___ (5)

∴ 4 - 5

→ $a(\alpha + \beta) + b - a(\beta + r) - b = 0$

→ $a((\alpha + \beta) - (\beta + r)) = 0$

→ $a(\alpha + \beta - \beta - r) = 0$

→ $a(\alpha - r) = 0$ ___ (6)

↳ Distinct value

→ $a = 0$

∴ Now, If $a = 0$ (6) then b should be "0" {from Eq^n 5} and if we Substitute (Eq 6 and 5) in {1} then c will be "0" as well.

∴ Then Our Quadratic Eqⁿ will become.

$0(x^2)+0(x)+0 \Rightarrow 0$ {for Eqⁿ 1, 6 and 5}

Note ⇒ This means that not only 1, Can place x value as α, β, γ but it Can be anything not only α, β and γ, it can be m, n, q, z etc.

⇒ So in Identity, $a \Rightarrow 0$, $b \Rightarrow 0$, $c \Rightarrow 0$ {always that's When we Can Call it an Identity property} this is only true if the question mention an eqⁿ quadratic has more than 3 roots.

Ex) Find the value of "a" for which the following eqⁿ has more than two roots,

$(a^2-5a+6)x^2-(a^2-3a+2)x+2a-a^2 \Rightarrow 0$

⇒ We know that this is an Identity property (cause it has more than 3 roots, which mean eqⁿ $ax^2+bx+c \Rightarrow 0$ {where $a \Rightarrow 0$, $b \Rightarrow 0$ and $c \Rightarrow 0$}

$(a^2-5a+6)x^2-(a^2-3a+2)x+2a-a^2 \Rightarrow 0$

$\downarrow\downarrow$ 0 $\quad\downarrow$ 0 $\quad\downarrow\downarrow$ 0

→ $a^2-5a+6 \Rightarrow 0$ → $a^2-3a+2 \Rightarrow 0$ → $2a-a^2 \Rightarrow 0$

→ $a^2-3a-2a+6 \Rightarrow 0$ → $a^2-2a-a+2 \Rightarrow 0$ → $a(2-a) \Rightarrow 0$

→ $a(a-3)-2(a-3) \Rightarrow 0$ → $a(a-2)-(a-2) \Rightarrow 0$ → $a \Rightarrow 0$ and $a \Rightarrow 2$

→ $(a-3)(a-2) \Rightarrow 0$ → $(a-1)(a-2) \Rightarrow 0$

⇒ $a \Rightarrow 3$ and 2 → $a \Rightarrow 1$ and 2

Values of a are ⇒ 0, 1, 2, 3.

Ex) Show that,

$$\frac{(x-a)(x-b)}{(c-a)(c-b)} + \frac{(x-b)(x-c)}{(a-b)(a-c)} + \frac{(x-c)(x-a)}{(b-c)(b-a)} \Rightarrow 1,$$

is an Identity.

Note: Roots,
We don't Call Something Roots Just Cause the output becomes 0. We Call them Roots Cause L.H.S = R.H.S {they Satisfy it}

→ $$\frac{(x-a)(x-b)}{(c-a)(c-b)} + \frac{(x-b)(x-c)}{(a-b)(a-c)} + \frac{(x-c)(x-a)}{(b-c)(b-a)} \Rightarrow 1$$

∴ We know for an Eq^n to be Identity it Should have 3 roots. then we Can Say it an Identity Eq^n. So, we must find 3 roots for this eq^n.

Suppose,

$\Rightarrow x \Rightarrow a$

$$\frac{(a-a)(a-b)}{(c-a)(c-b)} \to 0 + \frac{(a-b)(a-c)}{(a-b)(a-c)} \to 1 + \frac{(a-c)(a-a)}{(b-c)(b-a)} \to 0 \Rightarrow 1$$

$$0 + 1 + 0 \Rightarrow 1$$

$$1 \Rightarrow 1$$

$x \Rightarrow a$, is a root.

$x \Rightarrow b$

$$\frac{(b-a)(b-b)}{(c-a)(c-b)} + \frac{(b-b)(b-c)}{(a-b)(a-c)} + \frac{(b-c)(b-a)}{(b-c)(b-a)} \Rightarrow 1$$

$(b-a)(b-b) \to 0$, $(b-b)(b-c) \to 0$, $(b-c)(b-a) \to 1$

$$0 + 0 + 1 \Rightarrow 1$$

$$1 \Leftrightarrow 1$$

$x \Rightarrow b$ is a root

$x \Rightarrow c$

$$\frac{(c-a)(c-b)}{(c-a)(c-b)} + \frac{(c-b)(c-c)}{(a-b)(a-c)} + \frac{(c-c)(c-a)}{(b-c)(b-a)} \Rightarrow 1$$

$(c-a)(c-b) \to 1$, $(c-b)(c-c) \to 0$, $(c-c)(c-a) \to 0$

$$1 + 0 + 0 \Rightarrow 1$$

$$1 \Leftrightarrow 1$$

$x \Rightarrow c$

So the Eq^n has more than 3 roots and We know if a Eq^n has more than 3 roots then it is a Identity.

Relation between Roots and Coefficients

- If α and β are the roots of the quadratic Eq^n $ax^2+bx+c \Rightarrow 0$. then:

→ $ax^2+bx+c \Rightarrow a(x-\alpha)(x-\beta)$

multiply and divide by a so that we can make the coefficient 1.

$\frac{a}{a}x^2+\frac{b}{a}x+\frac{c}{a} \Rightarrow a(x-\alpha)(x-\beta)$

$x^2+\left(\frac{b}{a}\right)x+\left(\frac{c}{a}\right) \Rightarrow x^2-(\alpha+\beta)x+(\alpha\beta)$

→ $\alpha+\beta \Rightarrow \frac{-b}{a}$ and $\alpha\beta \Rightarrow \frac{c}{a}$

→ Sum of Roots $\Rightarrow \frac{-b}{a}$ and product of Roots $\Rightarrow \frac{c}{a}$

→ $\frac{-b}{a}$

→ Difference of Root $\Rightarrow |\alpha-\beta|$.

proof;

$(\alpha-\beta)^2 \Rightarrow (\alpha+\beta)^2-4\alpha\beta$

$(\alpha-\beta)^2 \Rightarrow \left(\frac{-b}{a}\right)^2-4\left(\frac{c}{a}\right)$

$\alpha-\beta \Rightarrow \frac{b^2}{a^2}-\frac{4}{a} \Rightarrow \sqrt{\frac{b^2-4ac}{a^2}} \Rightarrow \frac{\sqrt{b^2-4ac}}{\sqrt{a^2}}$

$\alpha - \beta \Rightarrow \frac{\sqrt{}}{a}$

$|\alpha - \beta| \Rightarrow \frac{\sqrt{}}{a} \Rightarrow \frac{|D|}{|a|}$

$ax^2 - bx + c \Rightarrow 0$

Sum of Roots $\Rightarrow \frac{b}{a}$

Product of Roots $\Rightarrow \frac{c}{a}$

→ Sum of Roots $\Rightarrow \alpha + \beta \Rightarrow -\frac{b}{a}$

→ Product of Roots $\Rightarrow \alpha \cdot \beta \Rightarrow \frac{c}{a}$

→ Difference of Roots → $|\alpha - \beta| \Rightarrow \frac{D}{|a|}$

Ex) If α and β are roots of $ax^2 + bx + c \Rightarrow 0$, find the value of following in term of a, b, c.

1) $\frac{1}{\alpha^2} + \frac{1}{\beta^2} \Rightarrow \frac{b^2 - 2ca}{c^2}$

⇒ We know

Sum of roots $\Rightarrow \alpha + \beta \Rightarrow -\frac{b}{a}$

Product of roots $\Rightarrow \alpha\beta \Rightarrow \frac{c}{a}$

Difference $\Rightarrow |\alpha - \beta| \Rightarrow \frac{D}{|a|}$

→ $\frac{1}{\alpha^2} + \frac{1}{\beta^2} \Rightarrow \frac{\beta^2 + \alpha^2}{\alpha^2\beta^2} \rightarrow \frac{\alpha^2 + \beta^2}{\alpha^2\beta^2}$

$\{ a^2 + b \Rightarrow (a+b)^2 - 2ab \}$

→ $\frac{\alpha^2 + \beta^2}{(\alpha\beta)^2} \Rightarrow \frac{(\alpha+b)^2 - 2\alpha\beta}{(\alpha\beta)^2} \Rightarrow \frac{(-b/a)^2 - 2(c/a)}{(\frac{c}{a})^2}$

→ $\frac{\frac{b^2}{a^2} - \frac{2c}{a}}{c^2/a^2} \Rightarrow \frac{\frac{b^2 - 2ca}{a^2}}{c^2/a^2} \rightarrow \frac{b^2 - 2ca}{c^2}$

2) $(a\alpha+b)^{-2} + (a\beta+b)^{-2} \Rightarrow \underline{b^2 - 2ca}$

$\rightarrow \dfrac{1}{(a\alpha+b)^2} + \dfrac{1}{(a\beta+b)^2}$

$\rightarrow$ ~~$\dfrac{1}{(a\alpha)^2+(b)^2+2}$~~

$\rightarrow \dfrac{(a\beta+b)^2 + (a\alpha+b)^2}{(a\alpha+b)^2(a\beta+b)^2}$

$\rightarrow \alpha+\beta \Rightarrow \dfrac{-b}{a}$, $\alpha \Rightarrow \dfrac{-b}{a} - \beta \Rightarrow \dfrac{-b-a\beta}{a} \Rightarrow \dfrac{(b+a\beta)^2}{(a)^2}$

$\rightarrow \dfrac{(a\alpha+b)^2+(a\beta+b)^2}{(a\alpha+b)^2(a\beta+b)^2} \Rightarrow \dfrac{a^2\alpha^2+b^2+2a\alpha b + a^2\beta^2+b^2+2a\beta b}{(a\alpha+b)^2(a\beta+b)^2}$

$\rightarrow$ Long process

$\rightarrow a\alpha^2+b\alpha+c \Rightarrow 0$

$\rightarrow a\beta^2+b\beta+c \Rightarrow 0$

$\rightarrow a\alpha+b \Rightarrow \dfrac{-c}{\alpha}$

$\rightarrow a\beta+b \Rightarrow \dfrac{-c}{\beta}$

$\rightarrow \left(\dfrac{-c}{\alpha}\right)^{-2} + \left(\dfrac{-c}{\beta}\right)^{-2} \Rightarrow \dfrac{c^2}{\alpha^2} + \dfrac{c^2}{\beta^2}$

$\Rightarrow \dfrac{c^2\beta^2+c^2\alpha^2}{\alpha^2\beta^2}$

$\Rightarrow \dfrac{c^2(\beta^2+\alpha^2)}{(\alpha\beta)^2}$ ∴

$$(\alpha x+b)^{-2}+(\alpha\beta+b)^{-2} \Rightarrow \frac{c^2(\alpha^2+\beta^2)}{(\alpha\beta)^2}$$

$$\left\{ (\alpha+\beta)^2 \Rightarrow \alpha^2+\beta^2+2\alpha\beta \right\}$$

$$\Rightarrow \frac{c^2\left((\alpha+\beta)^2-2\alpha\beta\right)}{(\alpha\beta)^2} \Rightarrow c^2\frac{\left((-b/a)^2-2(c/a)\right)}{\left(\frac{c}{a}\right)^2}$$

$$\Rightarrow c^2\left[\frac{b^2}{a^2}\right]-\frac{c^2 2c}{a}$$

$$\Rightarrow \frac{\frac{c^2b^2-2c^3a}{a^2}}{\frac{c^2}{a^2}}$$

$$\Rightarrow \frac{c^2(b^2-2ca)}{c^2}$$

$$\Rightarrow b^2-2ca$$

Ex. If one root of the Eqⁿ $px^2 - 14x + 8 \Rightarrow 0$ is 6 times the other, then p is Equal to?

→ One root is 6 time the other

If 1 is α
The other is 6α

→ Roots of the Eqⁿ → $(\alpha, 6\alpha)$

$px^2 - 14x + 8 \Rightarrow 0$

$p \Rightarrow ?$

→ $\alpha + \beta \Rightarrow \{\text{Sum of roots}\} \Rightarrow \frac{-b}{a}$

→ $7\alpha \Rightarrow \frac{+14}{p} \Rightarrow$ Sum of roots. — (1)

○ Product of roots $\Rightarrow \alpha\beta \Rightarrow \frac{c}{a} \Rightarrow \frac{8}{p} \Rightarrow 6\alpha^2$ — (2)

→ Divide (2) from (1)

→ $\frac{6\alpha^2}{7\alpha} \Rightarrow \frac{8}{p} \times \frac{p}{14}$

→ $6\alpha \Rightarrow \frac{14}{8} \rightarrow \alpha \Rightarrow \frac{8 \times 14^2}{14 \times 6 \times 3} \Rightarrow \frac{2}{21}$ — (3)

→ Substitute (3) in (1)

$7\left(\frac{2}{21}\right) \Rightarrow \frac{14}{p}$

→ $p \Rightarrow \frac{3 \times 14}{2} \Rightarrow \underline{21}$ $\{\therefore p = 21\}$

$\Rightarrow \frac{6\alpha^2}{7\alpha} \Rightarrow \frac{8}{\rho} \times \frac{\rho}{14}$

$\Rightarrow \alpha \Rightarrow \frac{7 \times 8}{6 \times 14} \Rightarrow \frac{2}{3} \Rightarrow \alpha \Rightarrow \frac{2}{3}$ — (3)

$\Rightarrow$ Substitute the value of α in (1)

$7\left(\frac{2}{3}\right) \Rightarrow \frac{14}{\rho}$

$\rightarrow \rho \Rightarrow \frac{3 \times 14}{2 \times 7} \Rightarrow 3$

$\Rightarrow \{\therefore \rho \Rightarrow 3\}$

Ex) Let α and β be the roots of the $x^2 - 6x - 2 \Rightarrow 0$. With $\alpha > \beta$ If $a_n \Rightarrow \alpha^n - \beta^n$ for $n = 1$ then the value of

$\frac{a_{10} - 2a_8}{2a_9}$

$\Rightarrow a_n \Rightarrow \alpha^n - \beta^n$

$a_{10} \Rightarrow \alpha^{10} - \beta^{10}$

$a_8 \Rightarrow \alpha^8 - \beta^8$

$a_9 \Rightarrow \alpha^9 - \beta^9$

$\Rightarrow \frac{\alpha^{10} - \beta^{10} - 2(\alpha^8 - \beta^8)}{2(\alpha^9 - \beta^9)} \Rightarrow \frac{a_{10} - 2a_8}{2a_9}$

$\Rightarrow \frac{\alpha^{10} - \beta^{10} - 2\alpha^8 + 2\beta^8}{2(\alpha^9 - \beta^9)}$

$\Rightarrow \frac{(\alpha^{10} - 2\alpha^8) - (\beta^{10} - 2\beta^8)}{2(\alpha^9 - \beta^9)}$

$\Rightarrow \frac{(\alpha^8(\alpha^2 - 2)) - \beta^8(\beta^2 - 2)}{2(\alpha^9 - \beta^9)}$

$\{\rightarrow \alpha^2 - 2 \Rightarrow 6\alpha \{\alpha^2 - 6\alpha + 2 \Rightarrow 0\}$
$\rightarrow \beta^2 - 2 \Rightarrow 6\beta \{\beta^2 - 6\beta - 2 \Rightarrow 0\}$

$\rightarrow \frac{\alpha^8(6\alpha) - \beta^8(6\beta)}{2(\alpha^9 - \beta^9)} \Rightarrow \frac{6(\alpha^9 - \beta^9)}{2(\alpha^9 - \beta^9)}$

$\Rightarrow \underline{3}$

Ex) find k in the eqⁿ $5x^2 - kx + 1 \Rightarrow 0$, such that the difference b/w the roots of the equation is unity.

→ Difference of two roots

$|\alpha - \beta| \Rightarrow 1$ (unity)

$|\alpha - \beta| \Rightarrow \frac{D}{|a|}$

$|\alpha - \beta| \Rightarrow D \Rightarrow |a|$ {for the eqⁿ to satisfy}

$D \Rightarrow |a|$

$D \Rightarrow \sqrt{b^2 - 4ac}$

→ $\sqrt{k^2 - 4(5)(1)} \Rightarrow |5|$

$\Rightarrow k^2 - 4(5) \Rightarrow (|25|)^2$

$\Rightarrow k \Rightarrow \pm\sqrt{45}$

Ex. If c, d are the roots of the Equation $(x-a)(x-b) - k \Rightarrow 0$, then a, b are the roots of the eqⁿ.

→ $(x-a)(x-b) - k \Rightarrow (x-c)(x-d)$

$(x-c)(x-d) + k \Rightarrow (x-a)(x-k)$

→ $(x-a)(x-b) - k \Rightarrow a(x-c)(x-d)$ (x^2 coefficient; factorized format)

→ $x^2 \Rightarrow 1$

→ $(x-a)(x-b) \Rightarrow (x-c)(x-d) + k$

Note: α, β 2 roots

→ $ax^2+bx+c \Rightarrow a(x-\alpha)(x-\beta)$

factorized format

Format of Quadratic Equation

→ A Quadratic eq^n whose roots are α and β, is

$\{(x-\alpha)(x-\beta)=0\}$ {You can also multiply a but if x coefficient is 1 then it won't make sense}

i.e $x^2-(\text{Sum of roots})x+(\text{product of roots})=0$

{in other words}

Ex) If α, β are the roots of the Quadratic eq^n $ax^2+bx+c=0$ then find the quadratic Eq^n whose roots are $\{2\alpha, 2\beta\}$

→ We know Sum of Roots $\Rightarrow \frac{-b}{a}$

Product of Roots $\Rightarrow \frac{c}{a}$

$\Rightarrow 2\alpha+2\beta \Rightarrow \frac{-b}{a}$

$\alpha+\beta \Rightarrow \left\{\frac{-b}{2a}\right\} \times 2 \Rightarrow \frac{-2b}{a}$

$\Rightarrow 2(\alpha 2\beta) \Rightarrow \frac{4c}{a}$

{Simple Calculation But made silly mistake}

→ $x^2-\left(\frac{-2b}{2a}\right)x+\left(\frac{4c}{2a}\right)=0$

$ax^2+2bx+4c=0$ $\{\therefore ax^2+2bx+4c=0\}$

Note: Method ⇒ 2

(α, β) are roots of

$ax^2 + bx + c \Rightarrow 0$

$\alpha \rightarrow$ root

$\rightarrow a\alpha^2 + b\alpha + c \Rightarrow 0$

Now;

$x \Rightarrow 2\alpha$

$\alpha \Rightarrow \frac{x}{2}$

$\rightarrow a\left(\frac{x}{2}\right)^2 + b\left(\frac{x}{2}\right) + c \Rightarrow 0$

$\rightarrow ax^2 + 2bx + c \Rightarrow 0$ — Hence

This is Only applicable if the root are Symmetric {I.E $(2\alpha, 2\beta)(4\alpha, 4\beta)(3\alpha, 3\beta)$}

It Wont Work if Roots are not Equal {I.E $(2\alpha, 3\beta), (3\alpha, 2\beta), (4\alpha, 5\beta)$ $(5\alpha, 4\beta)$}

Same Original Roots + Same Transformation ⇒ Symmetric transformation of roots.